

JavaScript Weakness List & Practice Tasks

Current Level

Strong:

- let / const
 - arrow function
 - template string
 - object destructuring
 - spread operator
 - map
 - filter
 - find
 - basic reduce
 - class
 - inheritance
 - optional chaining
 - dot & bracket notation
-

Weakness 1: Advanced Reduce

Task 1

```
const numbers = [10, 20, 30, 40, 50];
```

Find:

- max value
- min value

Using reduce only.

Task 2

```
const words = ['I', 'love', 'JavaScript'];
```

Create:

```
I love JavaScript
```

Using reduce.

Task 3

```
const users = [  
  {name: 'Rahim', salary: 20000},  
  {name: 'Karim', salary: 30000},  
  {name: 'Hasan', salary: 40000}  
];
```

Convert into:

```
{  
  Rahim: 20000,  
  Karim: 30000,  
  Hasan: 40000  
}
```

Using reduce.

Weakness 2: Array Destructuring

Task 4

```
const numbers = [10, 20, 30, 40, 50];
```

Get:

- first
- second
- last

Using destructuring.

Task 5

```
const colors = ['red', 'green', 'blue', 'black', 'white'];
```

Get:

- first
- third
- fifth

Using destructuring.

Weakness 3: Optional Chaining

Task 6

```
const company = {  
  departments: {  
    web: {  
      frontend: {  
        lead: 'Karim'  
      }  
    }  
  }  
}
```

Access lead using:

- dot notation
 - bracket notation
 - mixed notation
 - optional chaining
-

Task 7

Create a deeply nested object yourself.

Access a missing property without crashing the program.

Weakness 4: this Keyword

Task 8

Create:

```
class BankAccount
```

Properties:

- owner
- balance

Methods:

- deposit()
- withdraw()
- checkBalance()

Balance must change using this.

Task 9

Create:

```
class Product
```

Properties:

- name
- price

Methods:

- increasePrice()
- decreasePrice()

Must update object data.

Weakness 5: Class Design

Task 10

Create:

```
class Student
```

Methods:

- attendClass()
- payFee()
- changeDepartment()

Methods should update data.

Task 11

Create:

```
class Employee
```

Methods:

- promote()
- increaseSalary()
- transferDepartment()

Methods should update object data.

Weakness 6: Complex Array Problems

Task 12

```
const products = [  
  {name: 'Laptop', price:50000, quantity:2},  
  {name: 'Phone', price:30000, quantity:3},  
  {name: 'Monitor', price:15000, quantity:1},  
  {name: 'Mouse', price:500, quantity:5}  
];
```

Find:

1. product names
 2. grand total
 3. average price
 4. cheapest product
 5. most expensive product
 6. total quantity
 7. quantity > 2 products
 8. names of quantity > 2 products
 9. price after 10% discount
 10. total discount amount
-

Weakness 7: Find vs Filter

Task 13

```
const users = [  
  {id:1,name:'A'},  
  {id:2,name:'B'},  
  {id:3,name:'C'},  
  {id:2,name:'D'}  
];
```

Find:

- first user with id = 2
- all users with id = 2

Explain why outputs are different.

Before Moving Deep Into API

Must Be Comfortable With:

✓ map

✓ filter

✓ find

✓ reduce

✓ object destructuring

✓ array destructuring

✓ optional chaining

✓ class

✓ inheritance

✓ this keyword

✓ object access patterns

Next Topics

1. JSON
2. Fetch API
3. Promise
4. Async/Await
5. DOM + API Projects
6. Local Storage